
3AL Tutorial 7 2024 Solutions

1. Consider the group $\mathbb{Z}_2$ (under addition) and the set $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. We define a map $*$: $\mathbb{Z}_2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$[0] * (x, y) = (x, y)$$

$$[1] * (x, y) = (y, x),$$

where $(x, y) \in \mathbb{R}^2$.

- (a) Prove that $*$ is a group action.
(b) Find all orbits, stabilisers and the fixed set of the group action.

(a) By the definition of $*$, the identity condition for a group action is satisfied. Let $(x, y) \in \mathbb{R}^2$, we then have

$$[0] * ([0] * (x, y)) = [0] * (x, y) = ([0] + [0]) * (x, y)$$

$$[1] * ([0] * (x, y)) = [1] * (x, y) = ([1] + [0]) * (x, y)$$

$$[0] * ([1] * (x, y)) = [0] * (y, x) = (y, x) = [1] * (x, y) = ([0] + [1]) * (x, y)$$

$$[1] * ([1] * (x, y)) = [1] * (y, x) = (x, y) = [0] * (x, y) = ([1] + [1]) * (x, y).$$

Thus compatibility is satisfied too.

- (b) Let $(x, y) \in \mathbb{R}^2$. By definition, the stabiliser of (x, y) is

$$S((x, y)) = \{g \in \mathbb{Z}_2 : g * (x, y) = (x, y)\}.$$

We have two cases:

- (i) If $x = y$, then $S((x, y)) = \mathbb{Z}_2$.
(ii) If $x \neq y$, then $S((x, y)) = \{[0]\}$.

On the other hand, the orbit of (x, y) is

$$\mathbb{Z}_2 * (x, y) = \{g * (x, y) : g \in \mathbb{Z}_2\} = \{(x, y), (y, x)\}.$$

We have $(x, y) \in \mathbb{R}_f^2$ if and only if $x = y$. Thus, the fixed set is $\mathbb{R}_f^2 = \{(x, x) : x \in \mathbb{R}\}$.

2. Let G be an abelian group and X a set. Suppose $*$ is a group action by G on X .

- (a) Prove that $\circ : (G \times G) \times X \rightarrow X$ defined by $(g_1, g_2) \circ x = g_1^{-1}g_2 * x$ is a group action by $G \times G$ on X .
- (b) Let $x \in X$ and let $S_*(x)$ denote the stabiliser of x under $*$ and $S_\circ(x)$ denote the stabiliser of x under $\circ$. Prove that $(a, b) \in S_\circ(x)$ if and only if $aS_*(x) = bS_*(x)$.

(a) We have $(1, 1) \circ x = 1 * x = x$ for all $x \in X$. For $(a, b), (c, d) \in G$ and $x \in X$, we have

$$\begin{aligned} (a, b) \circ ((c, d) \circ x) &= (a, b) \circ (c^{-1}d * x) = a^{-1}b * (c^{-1}d * x) = a^{-1}bc^{-1}d * x \\ &= (ac)^{-1}(bd) * x = (ac, bd) \circ x = ((a, b)(c, d)) \circ x. \end{aligned}$$

Thus $\circ$ is a group action.

(b) $S_\circ(x) = \{(a, b) \in G \times G : a^{-1}b * x = x\} = \{(a, b) \in G \times G : a^{-1}b \in S_*(x)\}.$

3. Let G be a group. Consider each of the following maps $*_i : G \times G \rightarrow G$. Prove whether each $*_i$ is a group action or not.

- (a) $g *_1 x = xg$
- (b) $g *_2 x = xg^{-1}$
- (c) $g *_3 x = x^{-1}g$

(a) The map $*_1$ fails the compatibility condition since $g' *_1 (g *_1 x) = xgg'$ but $(g'g) *_1 x = xg'g$ and it is not necessarily true that $xgg' = xg'g$ (since that would require $gg' = g'g$, but G is not necessarily abelian).

(b) The map $*_2$ is a group action. We have $1 *_2 x = x1^{-1} = x$ and for $g_1, g_2 \in G$, we have

$$g_1 *_2 (g_2 *_2 x) = g_1 *_2 xg_2^{-1} = xg_2^{-1}g_1^{-1} = x(g_1g_2)^{-1} = (g_1g_2) *_2 x.$$

(c) The map $*_3$ is not a group action since it is not necessarily true that $1 *_3 x = x$ as that would require $x = x^{-1}$ for all $x \in G$. This is not true in most groups, e.g. in $\mathbb{Z}_4$, $[3] + [3] = [2] \neq [0]$ and so $[3]$ is not its own inverse.

Some other maps to think about: $g * x = g^{-1}xg$, $g * x = gx^{-1}$, $g * x = xgx^{-1}$

4. Let G be a finite group acting on a set X . Prove that for all $x \in X$, we have $|G| = |S(x)||G * x|$.

By the Orbit-Stabiliser Theorem, we have $|G * x| = |G : S(x)|$. Since G is a finite group, we can use Lagrange's Theorem to get $|G : S(x)| = |G|/|S(x)|$. Then $|G * x| = |G|/|S(x)|$ and so $|G| = |S(x)||G * x|$, as desired.

5. Let G be a group acting on a finite set X where $|G| = p$ for some prime p with $p \nmid |X|$. Prove that X_f is nonempty.

From the Orbit Decomposition Theorem, we have

$$|X| = |X_f| + \sum_{i=1}^n |G : S(x_i)|.$$

where $i = 1, \dots, n$ enumerates the distinct nonsingleton orbits $G * x_i$ of X . By Lagrange's Theorem, each $|G : S(x_i)|$ divides the order of G , that is, p . So we have either $|G : S(x_i)| = 1$ or p . However, if $|G : S(x_i)| = 1$, then x_i has a singleton orbit, which contradicts our earlier assumption. Thus, we must have $|G : S(x_i)| = p$ for all i . Thus $\sum_{i=1}^n |G : S(x_i)| = np$. Since $p \nmid |X|$ but $p \mid np$, we must have that $|X_f| \neq 0$. Thus, X_f is nonempty.

6. Let G be a group acting on a set X . The **fixer** of a group action is a subset F of G defined by $F = \{g \in G : g * x = x \text{ for all } x \in X\}$.

- (a) Prove that $F \triangleleft G$.
- (b) Show that X is a G/F -set.
- (c) Prove that the fixer of the action from (b) is the trivial subgroup of G/F .

HINT: Use $*_0$ and F_0 to denote the action by G/F on X and its fixer. This will avoid confusion with the action $*$ by G on X and its fixer F .

(a) First we need to show that F is a subgroup of G . Let $a, b \in F$ and $x \in X$. We need to show that $(ab) * x = x$ and $a^{-1} * x = x$. We have

$$(ab) * x = a * (b * x) = a * x = x.$$

Moreover,

$$x = 1 * x = (a^{-1}a) * x = a^{-1} * (a * x) = a^{-1} * x.$$

Thus $ab, a^{-1} \in F$. You could also prove this by noticing that if $a, b \in F$, then $a, b \in S(x)$ for all $x \in X$ and since $S(x)$ is a subgroup of G , $ab^{-1} \in S(x)$ for all $x \in X$ and so $ab^{-1} \in F$.

Now to show that F is normal in G . Take $g \in G$ and $f \in F$. Then

$$(gfg^{-1}) * x = g * (f * (g^{-1} * x)) = g * (g^{-1} * x) = gg^{-1} * x = 1 * x = x.$$

Thus $gfg^{-1} \in gFg^{-1}$ and so $F \triangleleft G$.

(b) We need to define an action by G/F on X . Elements of G/F have the form gF for $g \in G$. Consider the map $*_0 : G/F \times X \rightarrow X$ defined by $gF *_0 x = g * x$. We use a different notation for the group action because $*$ has already been used for the group action by G on X .

Because we are dealing with cosets, we need to show that the map is well-defined. That is, regardless of the choice of the coset's representative, the map behaves the same. Suppose $gF = g'F$. Then $g \in g'F$ and so $g = g'f$ for some $f \in F$. Thus

$$gF *_0 x = g * x = g'f * x = g' * (f * x) = g' * x = g'F *_0 x.$$

Therefore, $*_0$ is well-defined. It remains to show that $*_0$ satisfies the group action axioms. The identity of G/F is $F = 1F$ and we have

$$1F *_0 x = 1 * x = x.$$

Moreover, for $gF, g'F \in G/F$, we have

$$(gFg'F) *_0 x = gg'F *_0 x = gg' * x = g * (g' * x) = g * (g'F *_0 x) = gF *_0 (g'F *_0 x).$$

Thus there is a group action by G/F on X and so X is a G/F -set.

(c) Let F_0 be the fixer of $*_0$. We need to show that the only element of F_0 is the identity of G/F . That is, $F_0 = \{F\}$. Note that $gF \in F_0$ if and only if, for all $x \in X$,

$$x = gF *_0 x = g * x.$$

That is, $g \in F$, meaning $gF = F$. Therefore F is the only element of F_0 .